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A polymer physics model of the interphase cell nucleus for radiobiological simulations

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Advances in chromatin architectures achieved through super-resolution imaging and high-throughput chromosome conformation capture (Hi-C) techniques remain to be integrated into the modeling framework of radiobiology. In this study, the chromosomal and chromatin interactions in an interphase cell nucleus were described by polymer physics principles. To overcome the prohibitive computational cost, a multi-stage relaxation strategy was employed to decouple the relaxation processes of different structural levels. A distance-dependent DNA end rejoining model and a graph theory-based connected component analysis algorithm were implemented to simulate chromosome aberrations. Experimental measurements of chromosome aberrations in human skin fibroblasts exposed to both γ -rays and α particles were selected to benchmark the performance. The model efficiently reproduced 3D chromatin architectures, including chromosome territories and subcompartments, chromatin domains and loops. The predicted contact map and contact probability were consistent with Hi-C measurements. Model predicted dicentrics, interstitial deletions, and total aberrations aligned with experimental data within 20% for γ -rays and α particles, showing substantial improvement over previous reports. The proposed polymer physics-based model bridges structural biology and radiobiological simulations, demonstrating strong potential for accurate prediction of DNA damage and chromosome aberrations. Future refinements will integrate spatio-temporal repair dynamics and whole-genome sequencing data to enhance high-LET radiobiological effect predictions.

Understanding mechanisms of ionizing radiation induced proliferative cell death is critical for dose fractionation and relative biological effectiveness (RBE) estimation in particle therapy or space radiation^{1,2}. Among the various pathways that regulate radiation induced proliferative cell death, DNA damage and repair has been suggested to play an important role in the clinically relevant dose region³. Cells entering the cell cycle with unrepaired or misrepaired DNA damage may lose genes that are essential for cellular viability or undergo mitotic catastrophe. Radiobiological simulations based on Monte Carlo track structures and multi-level DNA architectures (the 2 nm DNA double-helix, the 10 nm nucleosome, and the 30 nm or above higher-order chromatin structures) have been successful in predicting the yield of DNA damage, especially DNA double-strand breaks (DSBs)^{4,5}. Mammalian cells have evolved strong DNA damage response mechanisms to antagonize isolated DSBs. However, if multiple DSBs are present in proximity (DSB cluster), heterologous DNA ends may be connected through the non-homologous end joining (NHEJ) or other repair processes thus producing chromosome aberrations (e.g. dicentrics, rings and deletions), which can be lethal to cells⁶. Two observations support the idea that DSB clusters are related to lethal damage. The first is that yields of dicentrics and cell survival curves show linear-quadratic dependence on radiation dose for X-rays, suggesting the interaction of multiple DSBs in the formation of lethal lesion and cell death⁷. The second evidence is that heavy ions, which cause dense ionizations and DSB clusters along the track, show less shoulder effect or approximately linear dose dependence for yields of dicentrics and cell survival⁸. Therefore, it is not only the sheer number of DSBs produced, but also the spatial distribution of DSBs within the cell nucleus and the subsequent interplay of DNA repair that play a critical role in determining the radiobiological effects.

To accurately simulate the spatial distribution of DSBs and the production of chromosome aberrations, one needs to establish models that can accurately reproduce the multi-level structures of the nucleus DNA, especially the packing of the 30 nm chromatin fiber in the interphase cell nucleus. Experimental observations using chromosome conformation capture and high-throughput sequencing (Hi-C) techniques have revealed

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that chromosomes are partitioned into subcompartments on the order of tens of Mb and further packed as topologically associated domains (TADs). Later improvement of the assay resolution down to 1 kb using in-situ Hi-C identified that TADs are actually chromatin contact domains and chromatin loops with median length of ~185 kb^{9–11}. Simulations of chromatin dynamics based on polymer physics principles have been successful in reproducing the realistic organization of chromatin. But they are limited to small systems, typically on the order of tens of Mbp DNA^{12,13}. For a cell nucleus system with DNA content on the order several Gbp, the computation workload can be extremely high. This kind of simulations is usually infeasible in a typical biophysics laboratory setting. Therefore, existing radiobiological simulation packages have taken alternative strategies by using either simplified chromosome architectures or predefined loop structures that largely overlooked the chromatin higher order structures and the chromatin interactions^{14–16}. The failure to employ a more realistic cell nucleus model may be a major source for the notable difference regarding the production of chromosome aberrations between experimental measurements and radiobiological simulations reported so far, especially for heavy ions^{17–20}. Ingram et al. leverage raw Hi-C contact data to iteratively refine nucleus geometries, with the primary goal of recapitulating experimental contact patterns and coupling these structures to radiation track simulations²¹. Their work provides critical insights into how Hi-C-derived chromosomal intermingling influences DNA damage distribution.

In this study, we leveraged the multi-level structures of chromatin to develop a multi-stage relaxation strategy for the generation of an interphase cell nucleus model based on polymer physics principles. In combination with other acceleration techniques, such as hierarchical searching and parallel computing, we reduced the computational workload by three orders of magnitude. We demonstrate that it is feasible to perform simulations of radiation induced DNA damage using the dynamically generated cell nucleus model based on polymer physics principles on a normal workstation computer. Moreover, by employing a distance-dependent DNA end random rejoining model with only one free parameter and an undirected graph based connected component analysis method, we are able to calculate the production of chromosome aberrations. Model predictions on different subtypes of chromosome aberrations for both γ -rays and α particles show substantial improvement over previous reports, suggesting great potential of radiobiological simulations using a polymer physics-based cell nucleus model.

Materials and methods

Polymer physics algorithms with multi-stage relaxation

The generation of the interphase nucleus DNA based on polymer physics principles was divided into three stages. In the first stage, chromosomes were simplified as polymers with a unit size (σ_{TAD}) comparable to that of a TAD to generate the distribution of TADs within the nucleus. A polymer unit (TAD) was subject to repulsions from all nearby units (distance $r < \sigma_{\text{TAD}}$) in any polymers in the system, tensions from adjacent units in the same polymer ($r > \sigma_{\text{TAD}}$), and the aggregation from the subcompartment. By assuming some basic forms of interactions (see Supplementary Data 1.1 for details), the force acting on a TAD unit in a cell nucleus can be written as

$$F_{\text{TAD}} = \sum_{i=1}^N \sum_{j=1}^{M_i} a \left(\sigma_{\text{TAD}} / r_{ij} \right)^2 - \sum_{i=1}^{i+1} k (r_i - \sigma_{\text{TAD}}) / \sigma_{\text{TAD}} - g \quad (1)$$

where a is the repulsive coefficient, k is the tensile coefficient, and g is the subcompartment cohesion constant. For the rest, N is the number of chromosomes in the nucleus, M_i is the number of TADs in the i_{th} chromosome, $\sigma_{\text{TAD}} = 300$ nm is the typical diameter of a TAD (estimated based on the genomic length of contact domains in relation to the genome size and cell nucleus volume, see Sect. 2 in Supplementary Data for details), r is the distance between two TAD centers. During the relaxation of the first stage, the positions of all the TADs were updated in each iteration with the displacement proportional to the force. A maximum step size of 10 nm and a random thermal motion of 3 nm were adopted to improve the stability of the convergence. The convergence was robust against reasonable variations in step size and random motion (Figure S1). A mirror reflection mechanism was applied to the cell nucleus membrane to stop TADs from crossing the nucleus boundary. Convergence was assumed if the total energy of the system did not decrease for 1000 iterations or the maximum iteration of 10,000 was reached.

Polymer physics algorithms for the generation of chromatin fiber distributions

Once the final positions of all the TADs in all the chromosomes were obtained, the second stage of the relaxation was initiated to generate the intermediate distributions of the chromatin fibers within each TAD separately. In this stage, the chromosomes were modeled as polymers with a unit size (σ_{CU}) comparable to that of the chromatin fiber unit. Besides repulsions and tensions similar to that of TADs, chromatin fiber units were subject to bending stress as well due to the stiffness of the fiber. In case there was a CTCF binding motif attached to the chromatin fiber, the cohesion of a pair of binding protein should also be considered. Taken together (see Supplementary Data 1.2 for details), the force acting on a chromatin fiber unit can be written as

$$F_{\text{CU}} = \sum_{i=1}^N \sum_{j=1}^{M_i} \sum_{k=1}^{K_{ij}} a \left(\sigma_{\text{CU}} / r_{ijk} \right)^2 - \sum_{i=1}^{i+1} k (r_i - \sigma_{\text{CU}}) / \sigma_{\text{CU}} - \gamma \sin \theta - b e^{(r/\sigma_{\text{CU}}-1)} \quad (2)$$

where $\gamma = Lp/\sigma_{\text{CU}}$ is the stiffness of the chromatin fiber, θ is the bending angle, b is the affinity of the binding protein, K_{ij} is the number of chromatin fiber units in the M_i th TAD, and $\sigma_{\text{CU}} = 32$ nm is the typical size of a

chromatin fiber unit. The persistence length L_p was taken as 85 nm based on experimental observations and Monte Carlo simulations²². The range of repulsion increased to $2\sigma_{CU}$ to reflect the extrusion effect of chromatin loops²³. Heuristic values of $a = 1.8$, $k = 20$ for the chromatin fiber and $b = 3$ for the binding protein were used to ensure that the distance between adjacent chromatin fiber units or the binding pairs were close to their physical dimension (32 nm). Chromatin loops existed in a part of TADs and they were assumed to tether at the TAD interfaces. In the second stage of the relaxation, the repulsions were only summed for units within the TAD rather than the entire nucleus to speed up the calculation. A maximum step size of 3 nm and a random thermal motion of 1 nm were adopted to improve the stability of the convergence. Variations of step size and random motion within reasonable range had minor influence on the convergence of the polymer system (Figure S2). A mirror reflection mechanism was applied on the boundary of TADs to hold the chromatin fiber units. The first and the last chromatin fiber units in a TAD were fixed at the TAD interfaces. Convergence was assumed for each TAD subsystem independently if the energy of the subsystem did not decrease for 100 iterations or the maximum iteration of 5000 was reached.

In the third stage of the relaxation, confinements of TADs were removed, and the nucleus boundary confinement was introduced back to the system. Chromatin fiber units were allowed to extrude and interact with any other nearby units in the nucleus. Due to the massive size of the system, each iteration would take a significant amount of time. Since the entire system was very close to the final steady state after the first two stages of relaxation, the final convergence could be assumed after a user-defined number of iterations were performed (500 iterations by default). The method described here has the flexibility to generate interphase cell nuclei of different shapes (e.g. sphere and ellipsoid) and DNA content (e.g. rat and human). Users are required to specify the dimension of the cell nucleus (X, Y, Z) in nm and the genomic length of each chromosome ($L_1, L_2, \dots, L_N$) in bp. The rest of the parameters were obtained from literature or fitting of relevant experiments and were hard coded in the algorithm to streamline the simulation workflow.

DNA damage scoring with hierarchical search and inverse transformation

Previously, we have developed a biophysical program to calculate ionizing radiation induced DNA damage using track structure data and a single chromosome model at the resolution of a nucleotide. Track structure data, which explicitly recorded the step-by-step interaction events of particles in the medium, was obtained using the open-source particle transport simulation code Geant4-DNA (version 4.10.7)²⁴. For the scoring of DNA damage, both the direct effect and the free radical-mediated indirect effect were modeled by overlaying track structures on cell nucleus models. Briefly, an ionization event located in the sugar-phosphate of the DNA backbone was assumed to create a single-strand break (SSB) and that in the nucleobase was assumed to result in a base damage. Ionization events located in the water medium had the potential to impact the nearby DNA double-helix through hydroxyls produced in radiolysis. The probability for a hydroxyl to create an SSB or base damage was modeled using a semi-empirical equation that incorporated the nonhomogeneous chemical reaction process and the cellular scavenging effect. A DSB was, by convention, defined as breaks (SSBs) on both sides of DNA strands within one helical turn (10 bp). The biophysical program was calibrated with the yield of SSBs for X-rays to obtain a few parameters. Model predictions were then extensively validated against experimental measurement in other scenarios. The overall good agreement with measured yields of SSBs, base damage, and DSBs for various types of radiation qualities proved the accuracy and generalizability of the established biophysical program. More detailed description of the model development and the validation of the model performance can be found in our previous works^{25,26}.

In this study, the DNA model was extended from a single chromosome to the entire genome in a cell nucleus, which brought about an increase in the DNA content by two orders of magnitude. To account for the sharp increase in the computer memory consumption and the computation time for DNA damage scoring, the hierarchical search and the inverse transformation mechanisms were introduced in this version of the biophysical program. Specifically, for each ionization event in the track structure file, to locate the nearest nucleotide out of the approximately 10^{10} nucleotides, the algorithm firstly identified the TADs (~268 kbp) within 500 nm of the ionization event (Fig. 1a). Then the algorithm searched for the nearest chromatin fiber unit (~2.3 kbp) within those TADs (Fig. 1b). Lastly, the algorithm located the nearest nucleotide within the chromatin fiber unit (Fig. 1c). This hierarchical search mechanism exploited the ordered organization of DNA and chromatin in the interphase nucleus, achieving very high efficiency. The inverse transformation mechanism was introduced to address the memory issue. In our previous work, the exact coordinates of all nucleotides in a chromosome were stored in the random-access memory (RAM), which usually took about 30 GB. However, to store the coordinates for all the chromosomes in a cell nucleus would take 2–3 TB of RAM, which was infeasible for a typical workstation computer. In this version of the program, we only loaded a chromatin fiber template and the coordinates and directions (in the format of 3D rotation matrix) of all the chromatin fiber units in RAM. When a specific chromatin fiber unit was called to search for the nearest nucleotide, an inverse transformation was performed to translate the ionization event to the local coordinate system of the chromatin fiber template (Fig. 1d). Mathematically, this was equivalent to the transformation of the chromatin fiber template to the global coordinate system of the ionization event but was much more computationally economic. In this way, the consumption of RAM was significantly reduced at almost no loss of computation efficiency. Figure 1e,f illustrate typical track structures and DSB distributions for γ -rays and of α particles, respectively.

A distance-dependent DNA end random rejoining model

To validate the polymer physics-based interphase cell nucleus model developed in this study, a distance-dependent non-homologous end joining (NHEJ) repair model was constructed (Fig. 2a). The basic assumption was that free DNA ends generated by DNA double-strand breaks could rejoin with any adjacent free DNA ends with certain probability. The probability was described by the formula $\exp(-r^2/2\delta^2)$, where r represented the

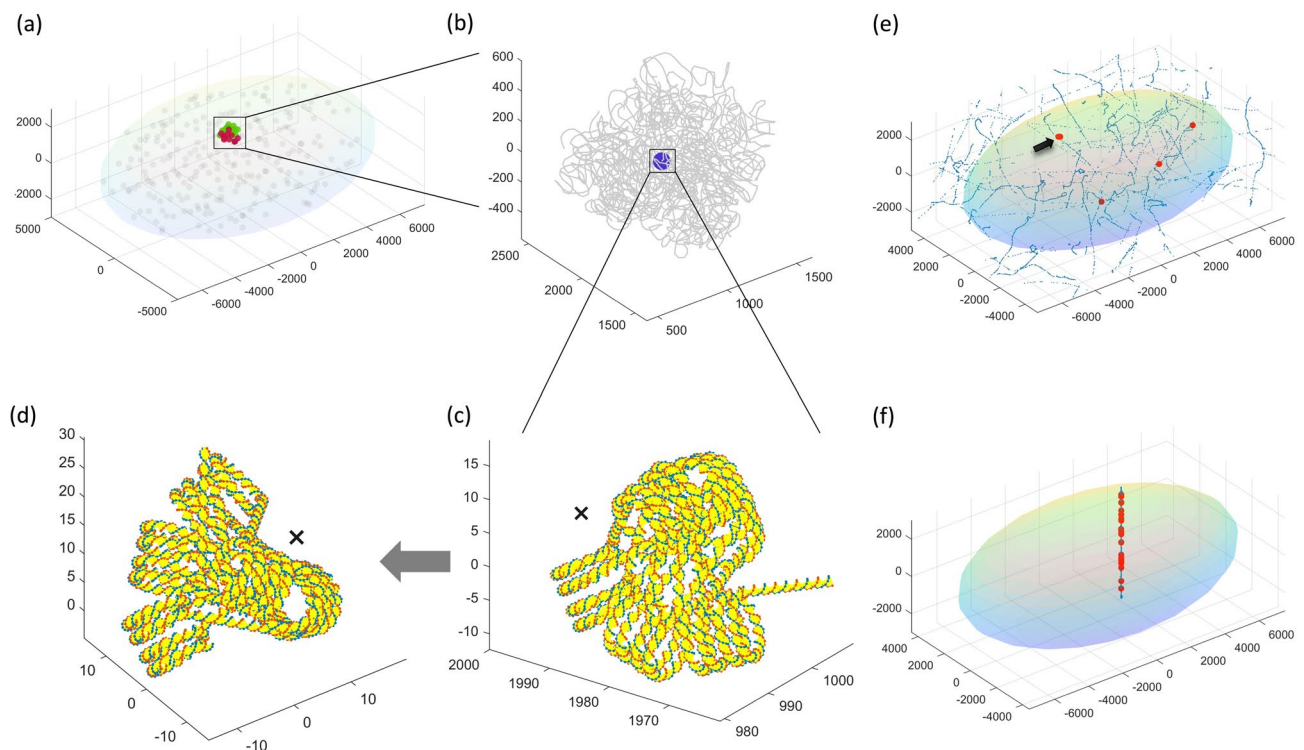


Figure 1. Schema of the hierarchical search and inverse transformation algorithm for DNA damage scoring. **(a)** Representative image of chromatin domains in relation to an ionization event. Chromatin domains within 500 nm of the ionization event are highlighted while the rest are visualized in light grey (only 5% of the domains are shown for clear view). Spatial coordinates are in nm for all figures. **(b)** Representative image of the distribution of chromatin fiber in relation to an ionization event. The blue dot indicates the location of the nearest chromatin fiber unit. **(c)** Visualization of a chromatin fiber unit at the resolution of single nucleotide in relation to an ionization event (marked as a cross) in the global coordinate system. **(d)** Visualization of a chromatin fiber template in relation to the ionization event in the local coordinate system. **(e,f)** Representative nano-scale energy deposition (blue dotted lines) and DSB distribution (red dots) patterns for γ -rays and α particles. The arrowhead indicates a DSB cluster for γ -rays.

distance between two DNA ends and δ was defined as the rejoining range constant, as proposed by McMahon et al.²⁷ The simulation of the repair process followed these steps. Firstly, a free DNA end was randomly selected from the pool of available free DNA ends. Then the model calculated the rejoining probability between this end and all other adjacent free ends within a predefined range. A target end was randomly chosen based on the probability distribution, and the two ends were marked as paired and rejoined. If no adjacent free ends existed within the range, the selected end was marked as unpaired. This process was repeated iteratively until all free DNA ends were processed. The maximum range for rejoining was set to 1 μm based on experimental observations of DNA repair foci: the diameter of fluorescent foci representing DNA repair sites was typically about 2 μm ²⁸, and the radius of these foci was adopted as the maximum rejoining range for DNA end pairing.

Analysis of chromosome aberrations based on graph theory

After running the DNA repair module, an undirected graph was constructed using misrepaired DNA ends and unrepaired DNA ends (left panel of Fig. 2b). All correctly rejoined DNA ends were removed from subsequent analysis. Specifically, the front and rear endpoints of all incorrectly repaired chromosomes were firstly listed as natural endpoints and assigned “telomere” attributes. Meanwhile, the upstream and downstream DNA ends of all incorrectly repaired DSBs were listed as break endpoints. Natural endpoints and break endpoints were taken together to serve as the vertices of the undirected graph. Next, the natural endpoints and DSB sites of each chromosome were sorted in order, and the DNA fragments formed between them were defined as natural edges. The “length” attribute of natural edges was set to the actual length of the DNA fragments, and the “centromere” attribute of natural edges denoted whether the corresponding DNA fragment contained a centromere. In addition, each pair of rejoined DNA ends was taken as a virtual repair edge, with the “length” attribute set to zero and no “centromere” attribute assigned. Natural edges and virtual repair edges were taken together to constitute all the edges of the undirected graph. Through the above procedures, an undirected graph was constructed to represent the reconnection schema of all chromosomes in the cell nucleus after radiation-induced DNA damage and repair.

Connected component analysis was then performed on the obtained undirected graph to determine the composition of new chromosomes and residual fragments (right panel of Fig. 2b). The Tarjan’s algorithm based

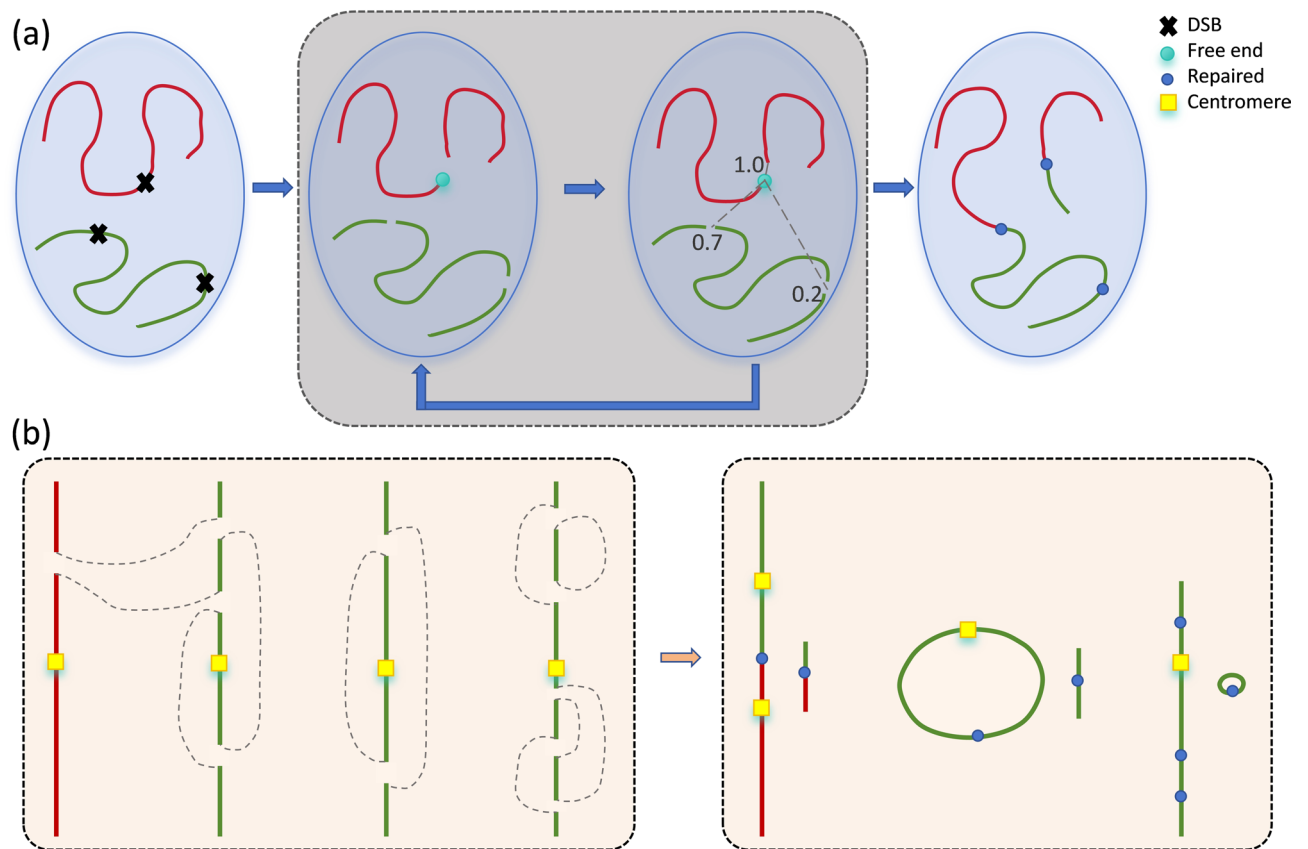


Figure 2. Schemata of the distance-dependent DNA end random rejoining model (a) and the undirected graph-based chromosome aberration scoring algorithm (b). Solid lines in red and green are simplified representation different chromosomes. Crosses represent initial DSB sites on chromosomes. Cyan dots represent DNA ends selected for the rejoining procedure, while blue dots indicate rejoining and fixation of two DNA ends. Locations of centromeres on chromosomes are marked as yellow squares.

on depth first search²⁹ was used and the implementation was validated on a range of test cases. The total length of a chromosome or fragment was determined by the sum of the lengths of all edges it contained. The number of centromeres was calculated as the total number of edges with “centromere” attributes, and the number of telomeres was determined by the total number of natural endpoints. Finally, the frequency of various types of chromosome aberrations was counted according to their biological definitions. Dicentric chromosomes were defined as chromosomes containing two centromeres; rings were defined as topologically closed chromosomes or fragments with head-to-tail connections; interstitial deletions were defined as removal of fragments inside chromosomes and were characterized by fragments without telomeres; and terminal deletions were defined as fragments containing one telomere. Since the resolution of G-banding karyotype analysis is typically 5–10 Mbp³⁰, a detection limit of $\epsilon = 10$ Mbp was adopted for deletion events when comparing simulated chromosome aberration frequencies with experimental measurements. Accordingly, the detection limit for rings was set to π to align simulation with experiment.

Specifications of the benchmark experiment and the simulation setup

The frequency of chromosome aberrations measured by Cornforth et al.³¹ was selected as the benchmark to evaluate the performance of the updated version of our biophysical program. The measurement was performed for both ¹³⁷Cs γ -rays (monoenergetic photons of 662 keV, absorbed dose 0–6.1 Gy) and ²³⁸Pu emitted α particles (estimated kinetic energy of 3.5 MeV as they traverse the nuclei, absorbed dose 0–2.2 Gy) using the primary human skin fibroblasts (AG1552) in density-inhibited G1 interphase. The four subtypes of chromosome aberrations, including dicentric chromosomes, rings, interstitial deletions, and terminal deletions, were scored for a total of 200–300 cells at each dose level for both γ -rays and α particles. The frequency of various types of chromosome aberrations was reported as a function of absorbed dose. Build-up regions of 4.5 μ m and 5 mm water layer were applied for α particles and γ -rays, respectively, to align with experimental setup. Interphase cell nuclei were constructed using the polymer-physics based algorithms with a cell dimension of [15 μ m, 10 μ m, 6 μ m] and DNA content of a human diploid cell (46 chromosomes, total DNA about 6 Gbp). The range of simulated radiation dose for γ -rays and α particles were selected as 1–7 Gy and 0.5–3 Gy, respectively. Dose of 0 Gy was not simulated in this study and the corresponding frequency of chromosome aberrations was taken as 0 assuming no spontaneous chromosome aberration in the biophysical program. For each type of radiation at

each radiation dose level, simulation was performed for 1000 cells (40 instances of the nucleus model combined with 25 radiation track structure files) to reduce statistical fluctuation. There was substantial variability in both the position and morphology of individual chromosomes, reflecting the strong randomness inherent in our geometry generation process (Figure S3 and S4). The chromatin fibers across the different model-generated nucleus geometries exhibited consistent spatial clustering characteristics as determined by the Ripley's K function (Figure S5). Since the real-time generation of an instance of the nucleus model was very time-consuming using the proposed method, a total of 40 instances of the nucleus model were prefabricated with random initial state and were re-used throughout the simulation study for efficiency consideration.

Results

Chromosome territories and gyration radius

The territory occupied by each chromosome and the distribution of TADs within each chromosome for an interphase cell nucleus were generated in the first stage of the relaxation. Figure 3a–c illustrate the highly compacted initial state, the intermediate state at 200 iterations, and the steady state of the 46 chromosomes in an ellipsoid cell nucleus. The convergence of the TAD system showed a fast initial decrease followed by a slow decrease as the system approached the steady state (Fig. 3d). The distances between adjacent TADs were mostly around 300 nm, although the distribution spectrum extended to 180 nm and 420 nm (Fig. 3e). Representative image of the chromosome territory, the chromosome subcompartment, and the distribution of TADs in three chromosomes is shown in Fig. 3f.

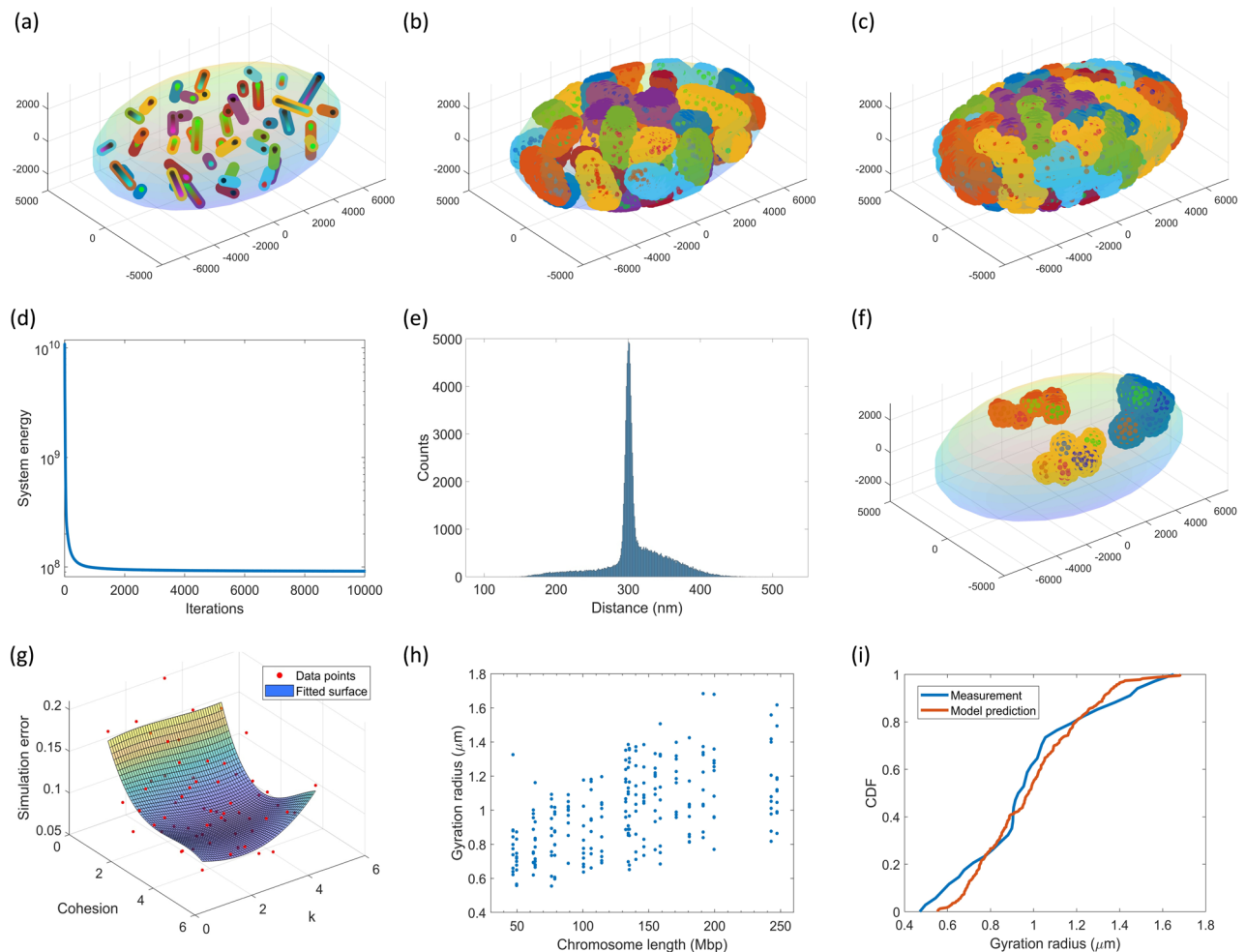


Figure 3. Chromosome territories and subcompartments in the stage one relaxation. (a–c) Representative images of chromosomes in a cell nucleus at the initial state, the intermediate state after 200 iterations, and the steady state, respectively. (d) Typical convergence curve of the stage one relaxation. (e) Distribution of distances between adjacent topologically associated chromatin domains. (f) Visualization of Chromosome territories and subcompartments for three chromosomes. (g) Surface plot of the simulation errors for gyration radius as a function of tensile coefficient and subcompartment cohesion constant. (h) Distribution of gyration radius for chromosomes with different genomic length. (i) Comparison of the cumulative density function for gyration radius between prediction and super-resolution imaging measurement.

There were three parameters in Eq. 1 that controlled the first stage relaxation. Among them, the repulsive coefficient a was normalized to unity while the tensile coefficient k and subcompartment cohesion constant g were obtained from fitting the simulated gyration radius of human cell in G1 phase to that of experimental measurement using super-resolution 3D single-molecule fluorescence microscopy (Fig. 3g)³². The combination of $k = 3.55$ and $g = 2.9$ that produced the least square error was adopted as the best-fit parameter set for TADs. The distribution of gyration radius in relation to the length of chromosomes in the simulation is provided in Fig. 3h. The cumulative density function for gyration radius predicted by the model agreed well with that of experimental measurements (Fig. 3i).

The distribution of interphase chromatin

The distribution of the chromatin fiber under the constraint of the virtual TAD boundary was produced in the second stage of the relaxation. Representative images of the initial state and the steady state of the chromatin fiber in a TAD are illustrated in Fig. 4a,b. Two types of TADs with loop domain and normal domain are illustrated in Fig. 4c. In some cases, two chromatin loops tethered to the same TAD interface, forming the so-called transitive looping (Fig. 4d)⁹. The current model setup results in approximately 9% of transitive looping, which is consistent with observed frequency of about 10%.

The final 3D architecture of the interphase chromatin of a human diploid cell nucleus demonstrated clear chromosome territory, chromosome subcompartment, and chromatin loop structures (Fig. 5a–c) as observed using the in situ Hi-C technique. The interaction of chromatin fiber in different TADs as a result of the third stage relaxation is illustrated in Fig. 5d,e. The decrease of energy of the entire system as a function of iteration numbers is shown in Fig. 5f. The pattern of the contact map of chromosome No. 10 again confirmed the formation of chromosome subcompartment and chromatin domain (Fig. 5g). The general trend and magnitude of the expected probability as a function of genomic distance for chromosome No. 10 measured with Hi-C was captured by our model prediction, although experimental measurement showed a smoother distribution with

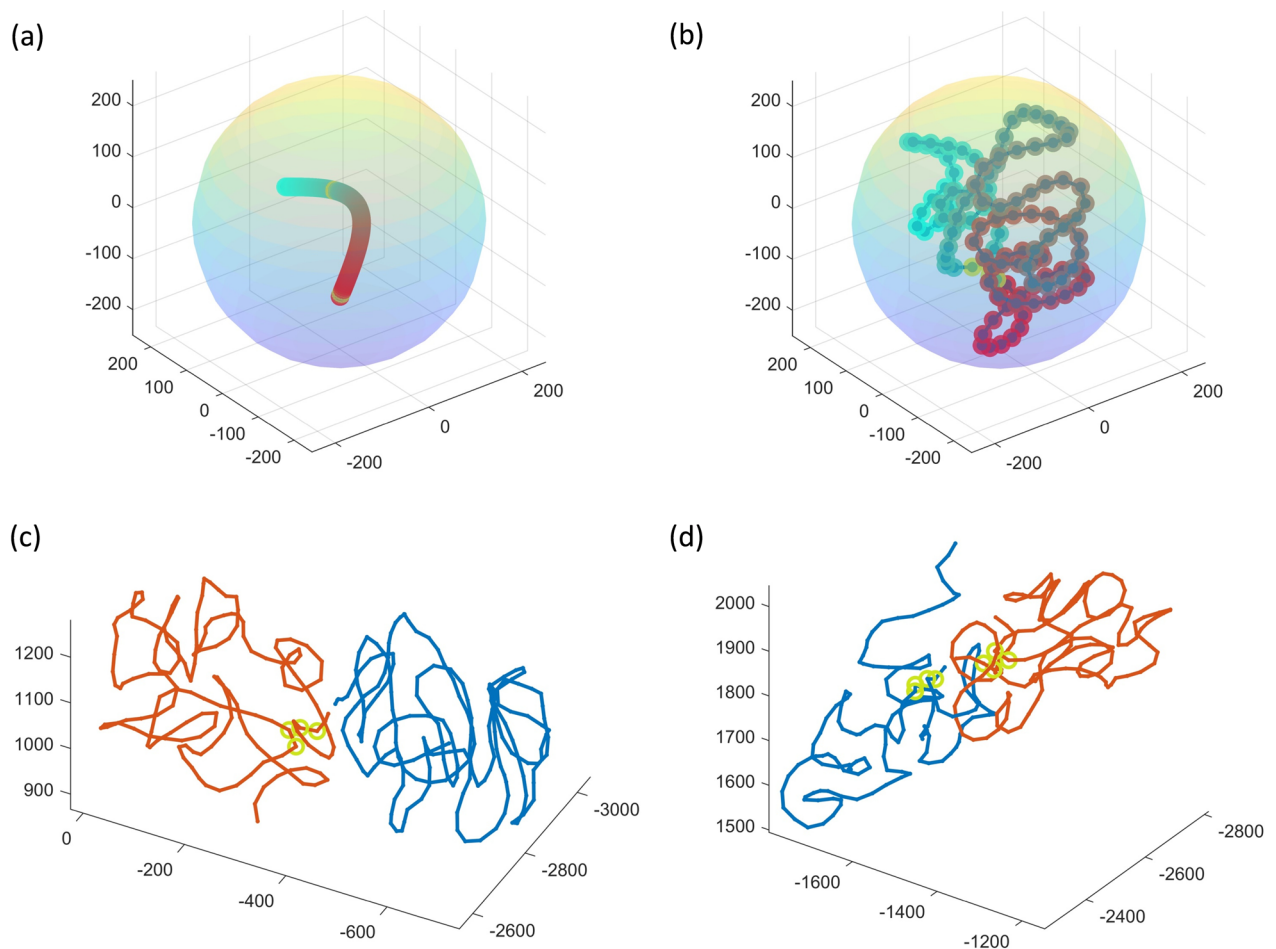


Figure 4. Topologically associated chromatin domains in the stage two relaxation. **(a,b)** Distribution of the chromatin fiber in a chromatin domain at the initial and steady state. **(c)** Representative image of a normal domain and a loop domain. The binding motifs are marked as yellow circles. **(d)** Representative image of transitive looping formed by two adjacent loop domains.

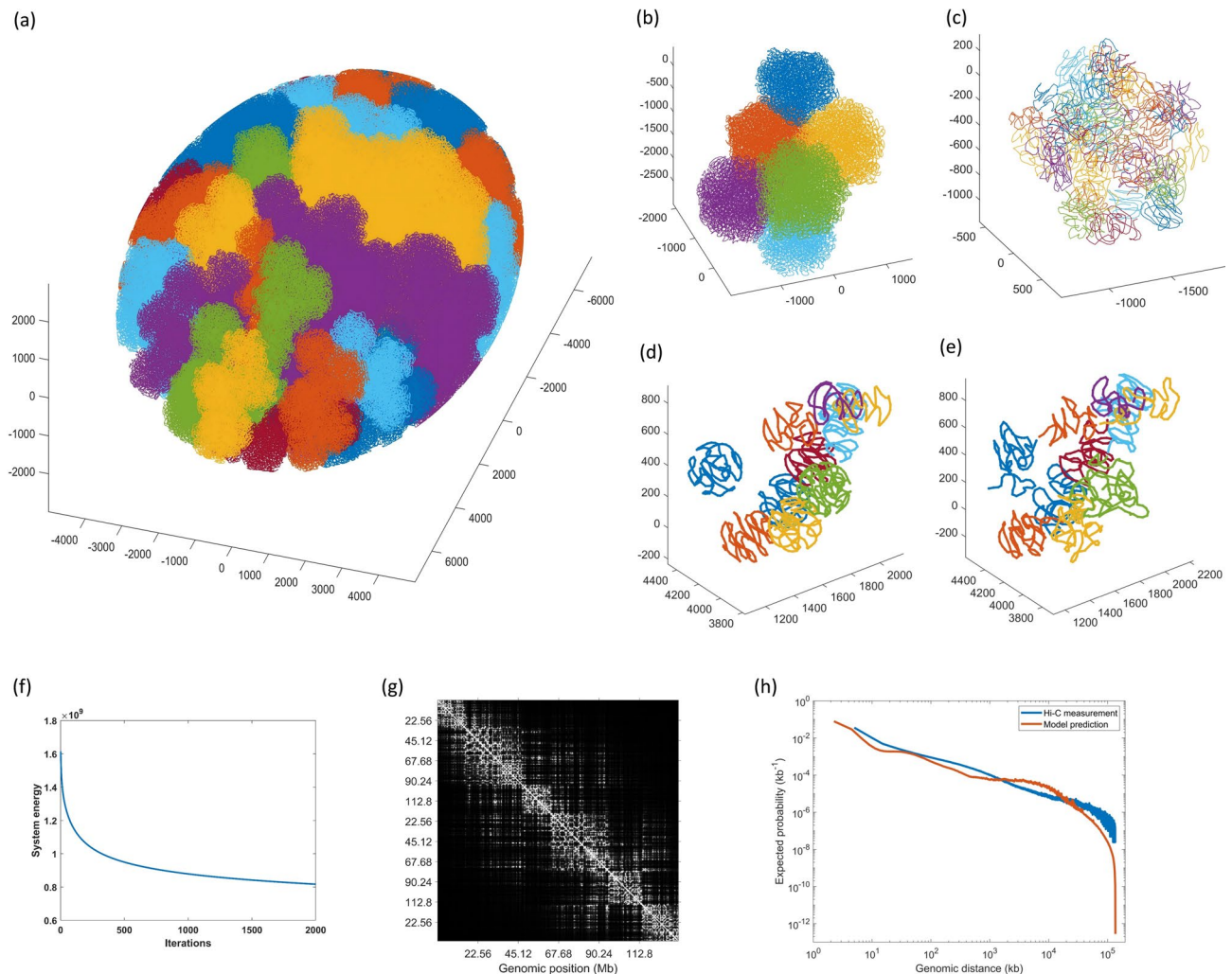


Figure 5. Architecture of the interphase chromatin in the stage three relaxation. **(a)** Visualization of the 3D chromatin architecture of an interphase cell nucleus for human fibroblast. **(b)** Visualization of a chromosome with different subcompartments shown in different colors. **(c)** Visualization of a subcompartment with TADs shown in different colors. Only half of the TADs are shown for clear view. **(d,e)** Representative images of chromatin domains before and after the stage three relaxation. **(f)** Typical convergence curve of the stage three relaxation for 2000 iterations. **(g)** Contact map for chromosome No. 10 with six subcompartments. **(h)** Comparison of the expected probability as a function of genomic distance for chromosome No. 10 between prediction and high-throughput chromosome conformation capture measurement.

less fluctuation (Fig. 5h). Other chromosomes showed similar trends and features (Figure S6). This suggests that the actual 3D architecture of the higher-order chromatin may be more complex and heterogeneous than modeled in this study. The root-mean-square error between simulation of measurement was 8.2×10^{-5} .

Efficiency and consistency

The first and the second stage of the relaxation took about 3700 s and 500 s, respectively, to reach convergence on a 64-core CPU with a base clock frequency of 3.1 GHz. The third stage of the relaxation took about 10 s per iteration. By default, the third stage relaxation was carried out for 500 iterations as a compromise of efficiency and accuracy based on the observed pattern of energy loss (Fig. 5f). Taken together, the generation of an interphase cell nucleus based on polymer-physics principles and the multi-stage relaxation strategy took about 10^4 s (3 h) in this study. Typically, it was estimated that it would take about 10^6 – 10^8 iterations for such a polymer system to converge without the multi-stage relaxation strategy^{33,34}. This corresponded to a total simulation time of about 10^7 – 10^9 s (4 months or more) on the same computer hardware as used in this study. Based on this, we estimated that the proposed model achieved a speedup factor of more than 1000.

To test the consistency of the biophysical program for the calculation of DNA damage, we performed simulations with different radiation qualities using the previous version (a single chromosome) and the recent version (the entire cell nucleus with inverse transformation for damage scoring) of the program. The calculated yields of DSBs for 1.25 MeV, 80 keV, and 1.5 keV photons and 5 MeV α particles were 7.2 ± 0.2 , 8.2 ± 0.3 , 12.4 ± 0.3 , and 15.1 ± 2.4 Gy⁻¹Gbp⁻¹, respectively, using the previous version. The uncertainty represented the standard

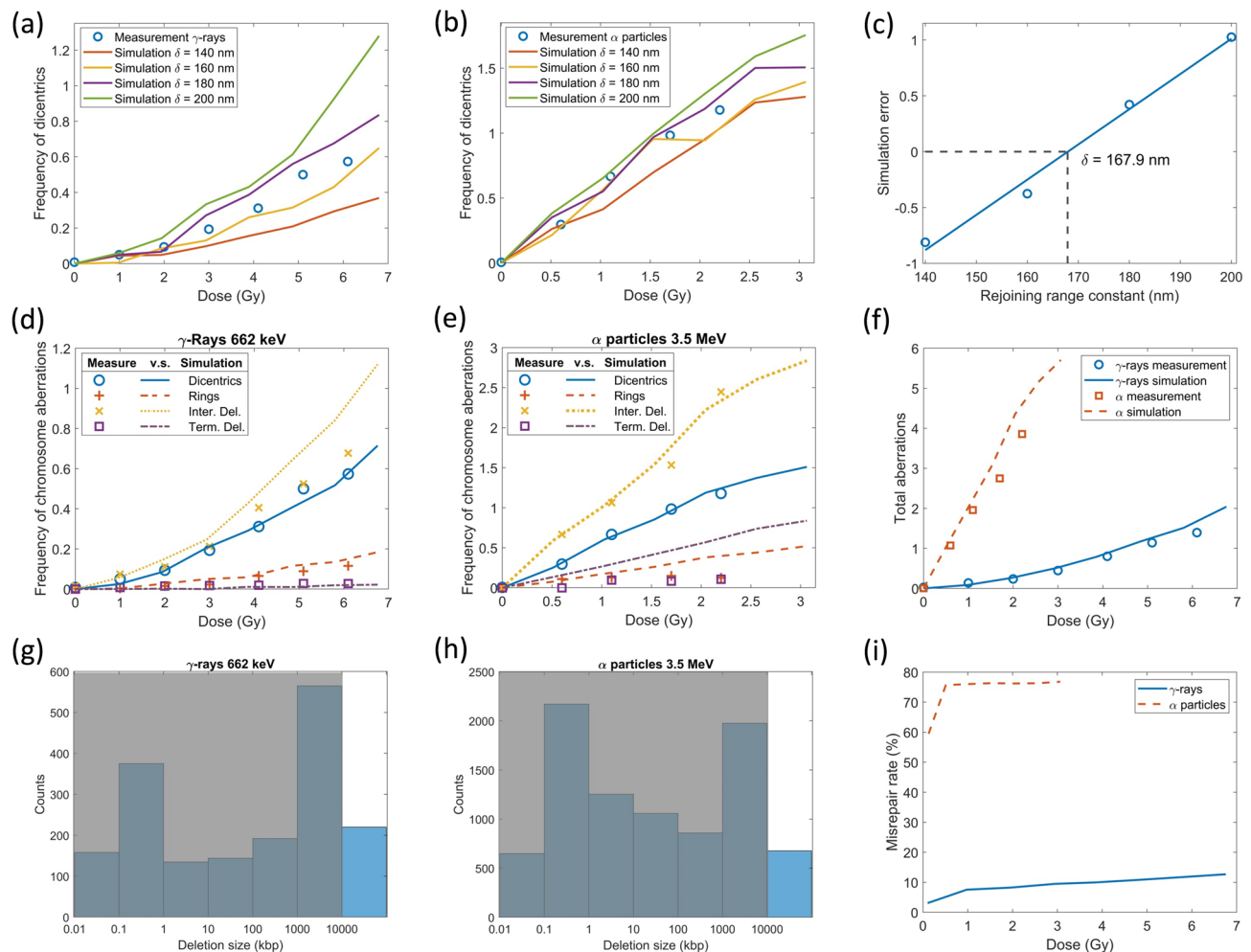


Figure 6. Model performance on the prediction of chromosome aberrations. (a,b) Simulated dependence of dicentric frequency with radiation dose for γ -rays and α particles under different rejoining range constants. (c) Linear fit of the simulation error for dicentric frequency in relation to the rejoining range constant for γ -rays. (d,e) Comparisons of model predicted frequency of dicentric, rings, interstitial deletions, and terminal deletions as a function of dose to that of measurement for γ -rays and of α particles. (f) Comparison of total aberrations between prediction and measurement. (g,h) Predicted spectra of deletion size for γ -rays and of α particles. Shaded areas mark deletion events below the G-banding detection threshold. (i) Model predicted ratio of misrepaired DSBs for γ -rays and α particles.

deviation of 5 repeat simulations. Consistent predictions were observed in the recent version with the yields of DSBs of 7.0 ± 0.2 , 8.5 ± 0.5 , 12.8 ± 0.7 , and $16.1 \pm 2.6 \text{ Gy}^{-1} \text{ Gbp}^{-1}$, respectively. Student's t-test suggest no statistical difference between the two groups of DSB predictions with P values of 0.21, 0.24, 0.37, and 0.52, respectively.

Prediction of chromosome aberrations

The simulated frequency of dicentric frequency as a function of dose for γ -rays was sensitive to the rejoining range constant δ , whereas the dependency was less sensitive for that of α particles (Fig. 6a,b). The measured frequency of dicentric frequency as a function of dose for γ -rays was used to fit the parameter δ . The best linear fit of $\delta = 167.9 \text{ nm}$ (Fig. 6c) was used in subsequent simulations. The model predicted frequency of the four types of chromosome aberrations as a function of dose for γ -rays and of α particles are presented in Fig. 6d,e, respectively, along with experimental measurement. The frequency of total aberrations for both γ -rays and of α particles are summarized in Fig. 6f. In general, predictions agreed reasonably well with measurement to within 20% for total aberrations. The predicted subtypes of chromosome aberrations mostly agreed well with measurement. However, the predicted rings and terminal deletions for α particles were ~ 3 and 6 times higher than those of measurement. The predicted spectra of deletion sizes for γ -rays and of α particles are shown in Fig. 6g,h, respectively. Our simulation results suggest that the majority of the deletion events are undetectable using the traditional G-banding technique (shaded area in Fig. 6g,h). Based on the prediction of the DNA repair module, the ratio of misrepaired DSBs was about 10% for γ -rays while that for α particles soared to more than 70% (Fig. 6i).

Discussion

The conversion of physical dose to equivalent biological dose is critical in particle therapy^{35,36}. The Microdosimetric-Kinetic model and the Local Effect Model are clinically established models based on the relation of “potentially lethal damage” or DSB to some form of “lethal damage”^{37,38}. In these models, the term “lethal damage” is conceptual, without a clear biological definition. Later models, such as the Repair-Misrepair-Fixation and the mechanistic model proposed by McMahon et al. assume that the formation of chromosome aberrations through the repair of DSBs is one kind of lethal damage^{27,39}. This assumption is reasonable since during mitosis, cells cannot appropriately divide dicentric chromosomes into their progenies. Besides, deletion or alteration of part of the DNA sequence in the genome may interfere with the proper functioning of essential genes for cell proliferation. Therefore, one substantial role of various radiobiological simulation codes or biophysical programs is to calculate/predict the frequency of chromosome aberrations for different radiation qualities. The PARTRAC code is one of the pioneers that extend the simulation of DNA damage to that of chromosome aberrations. The trend of the observed dependence of dicentrics with radiation dose is captured, although the model typically over-estimates the absolute yields of chromosome aberrations by a factor of 2–5^{15,17}. The code system developed by Nikjoo et al. has been extended to incorporate the DNA repair module. Deletions of small or intermediate size can be predicted through the error-prone repair pathways whereas large deletions are not interpretable in the model¹⁸. The Geant4-DNA and TOPAS-nBio toolkit developed on the Geant4 platform have also been upgraded to estimate DNA damage response^{19,24}. The RITCARD program, developed in collaboration with NASA, has been used to calculate chromosome aberrations for various ions (energies ranging from 10 to 1000 MeV/n), with over-estimation of simple exchange by a factor about 2 at high doses and better agreement at the lower dose region²⁰.

The accurate prediction of chromosome aberrations relies on the accurate prediction of the yield of DSBs and their distribution within the nucleus. The former issue has been addressed over the past decades with Monte Carlo track structure simulations and high-resolution DNA models. However, the latter, pertaining to the intricate higher-order chromatin architecture, still offers significant room for improvement. Current biophysical programs predominantly employ simplified configurations of chromatin in the cell nucleus using random walk, predefined loop structures, or Hilbert curves. The field of structural biology has witnessed great improvement for the understanding of the 3D architecture of the higher-order chromatin with in-situ Hi-C techniques^{9,40}. Unfortunately, these findings have not been fully integrated into the modeling framework in the radiobiological community. In this study, we closed the gap by implementing a polymer physics-based approach that incorporated the dynamic chromatin interaction in the formation of chromosome territories, chromosome subcompartments, and chromatin domains. Our analysis suggests that the proposed method successfully reproduces both the chromatin contact maps and the expected probability distributions of genomic interactions, aligning with experimental observations using Hi-C.

The inherent limitation of polymer physics-based modeling approaches lies in the prohibitive computational cost associated with relaxing such large-scale systems, which typically necessitates access to supercomputing facilities. This technical barrier has historically hindered the adoption of these approaches within the radiation biology community, where access to high-performance computing infrastructure is limited. In contrast, the multi-stage relaxation strategy developed in this study addressed this challenge through hierarchical spatial decomposition. The first stage focused on positioning TADs at micrometer-scale resolution, requiring approximately 3700 s to converge. The second stage then refined chromatin fiber organization within individual TADs, taking roughly 500 s for all domains. Finally, the third stage integrated the interaction of all these substructures under the nuclear boundary constraints, with each iteration taking about 10 s and converging within 500 iterations. The fixed iteration cap may lead to suboptimal chromatin distribution. Refinement (e.g. adaptive step-size) is in need to accelerate the third stage of relaxation. Collectively, this approach reduced the total simulation time to approximately 3 h on a standard 64-core workstation, representing a speedup factor of at least 1000-fold compared to conventional relaxation methods that would require 4 months or more. The implementation of inverse transformation further optimized computational efficiency during DNA damage scoring. By storing a chromatin fiber template and applying inverse coordinate transformations on-demand rather than precomputing all nucleotide positions, this method reduced RAM consumption from an infeasible 2–3 TB to manageable levels, while maintaining precise damage localization. Validation experiments confirmed that these optimizations do not compromise biological accuracy, as demonstrated by consistent DSB productions when compared to the previously established single-chromosome model. The combination of algorithmic innovations enables realistic whole-genome simulations of radiation-induced DNA damage in a typical radiation biophysics laboratory, thereby bridging the gap between structural biology insights and radiobiological modeling applications.

The critical role of chromosome aberrations in mediating proliferative cell death arises from their disruptive impact on genomic integrity. Despite advancements in track structure simulations and DNA damage modeling, current radiobiological programs have struggled to reliably predict chromosome aberration frequencies, as discussed previously. These limitations stem partly from oversimplified chromatin architectures. In contrast, our biophysical program addressed these shortcomings through three innovations: (1) a polymer physics-based nucleus model that reproduced TAD organization and chromatin domain dynamics, (2) a graph theory-based aberration analysis pipeline that accurately classified dicentrics, rings, and deletions using connected component analysis, and (3) a calibrated NHEJ repair model with distance-dependent rejoining probabilities. Validation against Cornforth et al.’s benchmark data³¹ demonstrated substantial improvement for γ -rays and α particles (agreed to within 20% of measured total aberrations), with particularly strong correlation for dicentrics and interstitial deletions. While terminal deletions for α particles remained elevated (~6 times higher than measurement), this discrepancy likely reflects the fixed maximum rejoining range assumption (1 μ m), which may require refinement for high-LET radiations to account for slower repair kinetics and larger diffusion

distances of complex damage foci. Collectively, these results establish our model as a robust tool for bridging radiation track structure simulations with clinically relevant biological endpoints.

With a resolution limit of 5–10 Mbp, conventional cytogenetic techniques fail to resolve the majority of small deletions predicted by our simulations (shaded regions in Fig. 6g,h), where approximately 90% of deletion events fall below the detection threshold. This technical limitation makes it difficult to properly interpret the experimental datasets. In contrast, whole-genome sequencing (WGS) offers sub-kilobase resolution capable of capturing the full spectrum of deletion sizes, which is critical for validating repair models^{41,42}. The NHEJ model used in this study was purely based on distance, which largely overlooked the repair protein dynamics. Meanwhile, other repair pathways, such as b-NHEJ, single-strand annealing, and homologous recombination, were not explicitly modeled. To address these limitations, future model iterations will integrate a spatio-temporal repair framework that dynamically simulate rejoining probabilities based on: (1) damage complexity (with longer time windows for clustered DSBs) and repair protein dynamics, (2) time-dependent diffusion coefficients for free DNA ends, and (3) cell cycle phase-specific repair dynamics and pathway choices. Additionally, coupling the repair model with WGS-derived deletion spectra will facilitate model validation. These advancements are critical for translating polymer physics-based nucleus models into relevant tools for particle therapy planning optimization or space radiation risk estimation, where accurate prediction of high-LET radiation effects remains paramount.

The current study has several limitations. (1) The expansion of the repulsive range to $2\sigma_{\text{CU}}$ to reflect loop extrusion was based on empirical considerations with reference to the morphological distribution of chromatin fibers (Figure S7), rather than strict physical or biological constraints. Although this parameter has minor impact on the total yields of aberrations, it can alter the distribution of aberrations between inter- and intra-chromosomal exchanges (Figure S8). (2) The concept of CTCF was solely adopted as a framework to introduce the description of chromatin loops. Numerous other factors including cohesin, co-factors, and additional architectural proteins have been demonstrated to contribute to chromatin loop regulation in living cells. (3) TADs were taken as primary units in this study whereas higher-order structures such as meta-TADs were not explicitly modeled. These features have been extensively reported in Hi-C and could influence long-range contact patterns and the expected probability curve^{43,44}. Moreover, recent advances in high-resolution techniques such as Micro-C have revealed more details on the fine structures of chromatin conformation^{45,46}. (4) In addition to the gyration radius benchmark experiment, other benchmark analyses such as the radial positioning of chromosomes and the distance distributions between FISH-labeled loci would provide a more comprehensive assessment of how well the model recapitulates native chromatin architecture^{47,48}. (5) The use of a combination of 40 distinct nucleus geometries and 25 unique radiation track structure files to generate 1000 simulated cell irradiations was primarily driven by practical feasibility constraints. This treatment may introduce a bias in the calculation of exchange probabilities between specific chromosomes.

Data availability

The datasets used in the current study are available from the corresponding author upon reasonable request.

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Author contributions

M. Z.: Software, Data curation, Formal analysis, Writing-original draft. G. H.: Methodology, Data curation, Writing-review and editing. Z. X.: Investigation, Writing-review and editing. X. H.: Supervision, writing-review and editing. Y. L.: Writing-original draft, Conceptualization, Resources.

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Competing interests

The authors declare no competing interests.

Additional information

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